

Hardware Sensor Test Documentation: Gyroscope

Module: Gyroscope Sensor Testing

Functionality: Angular Velocity & 3D Motion Tracking

Test Case ID	Description	Pre-conditions	Test Steps	Expected Result	Status
TC_GYRO_001	Verify real-time angular velocity measurement (Yaw, Pitch, Roll).	Sensor Test App or Factory Diagnostic Menu is open.	1. Rotate the device along the X-axis (Pitch). 2. Rotate along the Y-axis (Roll). 3. Rotate along the Z-axis (Yaw).	The sensor must dynamically detect angular changes in rad/s or deg/s across all three axes with zero visible latency.	Pass
TC_GYRO_002	Verify sensor stability when the device is completely static.	Device is placed flat on a vibration-free, solid desk.	1. Open a hardware sensor graph logger. 2. Monitor raw Gyroscope values for 30 seconds.	Drift values must remain strictly within nominal thresholds (approaching 0.00 rad/s), without erratic spikes.	Pass

Functionality: Gaming, AR View, & Camera Stabilization

Test Case ID	Description	Pre-conditions	Test Steps	Expected Result	Status
TC_GYRO_003	Verify motion-controlled steering accuracy in video games.	A racing or FPS game supporting Gyro-aiming is installed and configured.	1. Tilt the physical device slightly left and right during a gameplay session.	The game engine must map the device's tilt directly to vehicle steering or weapon aiming smoothly and precisely.	Pass
TC_GYRO_004	Verify 360-degree camera viewing and AR rendering.	An Augmented Reality (AR) or Google Maps Street View app is running.	1. Hold the device up and pan around in a 360-degree circle.	The virtual environment/camera viewfinder must shift accurately based on physical spatial rotation.	Pass
TC_GYRO_005	Verify Electronic Image Stabilization (EIS) during video recording.	Camera app is active. Video stabilization toggle is enabled.	1. Start a video recording. 2. Walk or move the hand rapidly up and down to create minor shakes.	The system must utilize real-time gyro data to apply algorithmic crop and stabilize the recorded video output cleanly.	Pass

Functionality: Advanced UI Gestures & Power Optimization

Test Case ID	Description	Pre-conditions	Test Steps	Expected Result	Status
TC_GYRO_006	Verify interaction with native Compass/Pedometer apps.	System Compass app is opened.	1. Rotate the device horizontally to calibrate. 2. Walk with the phone in hand.	The gyro must cross-reference data with the magnetometer to prevent compass lag and ensure micro-step count accuracy.	Pass
TC_GYRO_007	Verify sensor power state transition during Always-On Display (AOD).	AOD is enabled. Device is locked.	1. Place the device face up. 2. Monitor power consumption logs using a hardware profiler or ADB.	The gyroscope must enter a smart low-frequency polling mode or suspend state until a motion interrupt wakes the display.	Pass